

COURSE INFORMATION

School/Faculty:	Faculty of Computing		
Program name:	BSc in Computer Science (Computer Networks and Cyber Security) with honour		
Course code:	SCSC2023	Academic Session/Semester:	20262027/1
Course name:	Computer Architecture & Assembly Language	Pre/co requisite (course name and code, if applicable):	SCSC1013 Computer System & Organization
Credit hours:	3		

Course synopsis	This course introduces students to the principles of computer architecture and assembly language programming, bridging the gap between hardware and software systems. Students will explore topics such as CPU architecture and microarchitecture, instruction set architectures, floating-point representation and arithmetic, performance evaluation, and advanced architectural concepts. The course also covers assembly language programming using 32-bit x86 processors, including data transfer, addressing modes, arithmetic and logical operations, procedures, conditional execution, and bitwise manipulation. Through lectures, laboratory sessions, and project-based learning, students gain hands-on experience in writing low-level programs and using simulators to analyze and debug system behavior. By the end of the course , students will be able to analyze architectural designs, apply assembly programming principles, and collaboratively implement real-world solutions using structured problem-solving and low-level programming techniques.			
Course coordinator (if applicable)	Mohd Foad Bin Rohani, Dr.			
Course lecturer(s)	Name	Office	Contact no.	E-mail @utm.my
	Mohd Foad Bin Rohani, Dr.	N28-347-05		foad

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	* CLO	PLO ** (MQF Cluster Code)	*** Taxonomies and **** Graduate Attributes	T&L methods	***** Assessment methods
CLO1	Analyze the fundamental principles of computer architecture design by utilizing systems thinking to organize and integrate key components effectively.	PLO1 (C1)	C4 SC1	Lectures	Q, T
CLO2	Evaluate the fundamental principles of assembly language programming by using strategic thinking to integrate and organize key programming concepts effectively, including code quality, testing, and maintainability considerations.	PLO2 (C2)	C5 SC2	Lectures	Q, F
CLO3	Practice using simulation tools to design computer architecture components and debug assembly language programs, demonstrating integrated problem solving to enhance accuracy and effectiveness in simulation and program development, including modular design and structured testing practices.	PLO3 (C3A)	P3 SC3	Hands-on Lab	L
CLO4	Demonstrate an assembly language program that addresses practical problems or explores advanced concepts in computer architecture, using collaborative skills to enhance problem-solving through teamwork and shared insights, with appropriate code documentation and modular implementation.	PLO4 (C3B)	A4 A(S11)	Project-based Learning	PR

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.

* Up to 5 CLO

Refer *** Taxonomies of Learning and **** UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement

***** L- Lab; T – Test; Q – Quiz; PR – Project; F – Final Exam

** MQF Cluster Code

C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C = Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Entrepreneurial Skills, C5 = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Hands-on Lab	Implemented via structured lab sessions that emphasize practical application of concepts through tools, software, or hardware.
2	Project-based Learning	Students in a group of four (4) are assigned a project to design and implement assembly programming solution for a real-world problem by integrating theory and appropriate tools. They also need to evaluate the implementation of the project. The project is divided into several deliverables.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

This course develops transferable skills by engaging students in both theoretical and practical aspects of computer architecture and assembly language. Through lectures, hands-on labs, and group projects, students enhance their problem-solving, system thinking, and technical skills. Practical tasks using simulators and low-level programming help build attention to detail and precision. Collaborative project work improves teamwork and communication abilities as students work together to design and implement real-world solutions. These experiences also promote strategic thinking and the ability to apply structured approaches—skills that are useful across a range of academic and professional settings.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities										TOTAL SLT		
			Face-to-Face (F2F)								Non F2F Independent Learning				
			Physical				Online (Synchronous)				Online (Asynchronous)	Others			
			L	T	P	O	L	T	P	O					
Week 1	Introduction to Computer Architecture	CLO1					3						1	4	
Week 2	Floating Point Representation and Arithmetic 1	CLO1	2			1								1	4
Week 3	Floating Point Representation and Arithmetic 2	CLO1	2			1							1	2	6
Week 4	Instruction Set Architectures (ISAs) - RISC & CISC	CLO1	2			1								1	4
Week 5	CPU Architecture and Microarchitecture 1	CLO1	2			1								1	4
Week 6	CPU Architecture and Microarchitecture 2	CLO1, CLO3	2			1							1	2	6
Week 7	Performance Measurements and Analysis	CLO1, CLO4	2			1								1	4
Week 8	Semester Break														
Week 9	Introduction to 32-Bit x86 Processors (1W)	CLO2	2			1							1	2	6
Week 10	Assembly Language Fundamentals	CLO2	2			1							1	6	10
Week 11	Data Transfers, Addressing, and Arithmetic 1	CLO2, CLO3	2			1								1	4
Week 12	Data Transfers, Addressing, and Arithmetic 2	CLO2	2			1							1	2	6
Week 13	Procedures and Function Calls	CLO2, CLO3, CLO4	2			1								1	4
Week 14	Conditional Processing	CLO2, CLO4	2			1								1	4
Week 15	Boolean and Bitwise Operations	CLO2	2			1							1	2	6
SUB-TOTAL SLT :												72			

Continuous Assessment		%	Face-to-Face (F2F)		Non F2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Lab	15	6		3	6	15
2	Quiz	20	2			4	6
3	Test	20	2			4	6
4	Project	15	6		1	5	12
SUB-TOTAL SLT :							39

Final Assessment		%	Face-to-Face (F2F)		Non F2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	30	3			6	9
SUB-TOTAL SLT :							9
SLT for Assessment :							48
GRAND TOTAL SLT :							120

		GRAND TOTAL SLT :	120
A	% SLT for F2F Physical Component :		48.33
B	% SLT for Online & Independent Learning Component :		51.67
C	% SLT for Online Component :		10.83
D	% SLT for All Practical Component :		10.83
D1	% SLT for F2F Physical Practical Component :		10.83
D2	% SLT for F2F Online Practical Component :		0.00
Please tick (✓) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)			

Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)
Visual Studio 2019

References:

1. Elahi, A. (2022). Computer systems: Digital design, fundamentals of computer architecture and ARM assembly language. Springer. https://doi.org/10.1007/978-3-030-93449-1
2. Irvine, K. (2019). Assembly language for x86 processors (8th ed.). Pearson Education.
3. Ledin, J. (2022). Modern computer architecture and organization: Learn x86, ARM, and RISC-V architectures and the design of smartphones, PCs, and cloud servers. Packt Publishing.
4. Patterson, D. A., & Hennessy, J. L. (2011). Computer organization and design: The hardware/software interface (5th ed.). Morgan Kaufmann.
5. Stallings, W. (2019). Computer organization and architecture: Designing for performance (11th ed.). Pearson.

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such.
Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information:

All students must adhere to the following to pass a computing course:
i. Pass both Continuous Assessment AND Final Assessment.
ii. Get at least minimum pass marks following UTM regulation (**40% or D+**).

The passing mark for:

- Continuous Assessment is **30%** of continuous marks.
- Final Assessment is **20%** of the final marks.

If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark.

Disclaimer:

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$ELT = (\text{Theory} + \text{Industrial Guidance} + \text{Assessment}) \times 50\%$
Total of credit for LI/Practical = $ELT/40$ Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:		Certified by:	
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Signature	: 	Signature	: 
Date	: 25-Jan-2026	Date	: 25-Jan-2026